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Breast Cancer Staging

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TNM Classification for Breast Cancer

The American Joint Committee on Cancer (AJCC) provides two principal groups for breast cancer staging: anatomic, which is based on extent of cancer as defined by tumor size (T), lymph node status (N), and distant metastasis (M); and prognostic, which includes anatomic TNM plus tumor grade and the status of the biomarkers human epidermal growth factor receptor 2 (HER2), estrogen receptor (ER), and progesterone receptor (PR).

The prognostic stage group is preferred for patient care and is to be used for reporting of all cancer patients in the United States.[1,2,3] Ductal carcinoma in situ and stage I, stage IIA, stage IIB, and stage IIIA breast cancers are considered early-stage breast cancer (ie, cancer that has not spread beyond the breast or the axillary lymph nodes).[4]

In turn, prognostic stages are divided into clinical and pathological groups. Pathological stage applies to patients who have undergone surgery as the initial treatment for breast cancer. It includes all information used for clinical staging plus findings at surgery and pathological findings from surgical resection. Pathological prognostic stage does not apply to patients who received neoadjuvant therapy (systemic agents or radiation prior to surgical resection).[1,2,3,5] See the tables below.

Table 1. TNM Classification for Breast Cancer (Open Table in a new window)

Primary tumor (T)	
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	Carcinoma in situ
Tis (DCIS)	Ductal carcinoma in situ
Tis (Paget)	Paget disease of the nipple NOT associated with invasive carcinoma and/or carcinoma in situ (DCIS) in the underlying breast parenchyma. Carcinomas in the breast parenchyma associated with Paget disease are categorized on the basis of the size and characteristics of the parenchymal disease, although the presence of Paget disease should still be noted
T1	Tumor ≤ 20 mm in greatest dimension

T1mi	Tumor ≤ 1 mm in greatest dimension
T1a	Tumor > 1 mm but ≤ 5 mm in greatest dimension (round any measurement >1.0-1.9 mm to 2 mm)
T1b	Tumor > 5 mm but ≤ 10 mm in greatest dimension
T1c	Tumor > 10 mm but ≤ 20 mm in greatest dimension
T2	Tumor > 20 mm but ≤ 50 mm in greatest dimension
T3	Tumor > 50 mm in greatest dimension
T4	Tumor of any size with direct extension to the chest wall and/or to the skin (ulceration or skin nodules), not including invasion of dermis alone
T4a	Extension to chest wall, not including only pectoralis muscle adherence/invasion
T4b	Ulceration and/or ipsilateral satellite nodules and/or edema (including peau d'orange) of the skin, which do not meet the criteria for inflammatory carcinoma
T4c	Both T4a and T4b
T4d	Inflammatory carcinoma
Regional lymph nodes (N)	
Clinical	
cNX	Regional lymph nodes cannot be assessed (eg, previously removed)

cN0	No regional lymph node metastasis (on imaging or clinical examination)
cN1	Metastasis to movable ipsilateral level I, II axillary lymph node(s)
cN1mi	Micrometastases (approximately 200 cells, larger than 0.2 mm, but none larger than 2.0 mm)
cN2	Metastases in ipsilateral level I, II axillary lymph nodes that are clinically fixed or matted; or in ipsilateral internal mammary nodes in the absence of clinically evident axillary lymph node metastases
cN2a	Metastases in ipsilateral level I, II axillary lymph nodes fixed to one another (matted) or to other structures
cN2b	Metastases only in ipsilateral internal mammary nodes and in the absence of axillary lymph node metastases
cN3	Metastases in ipsilateral infraclavicular (level III axillary) lymph node(s), with or without level I, II axillary node involvement, or in ipsilateral internal mammary lymph node(s) with level I, II axillary lymph node metastasis; or metastases in ipsilateral supraclavicular lymph node(s), with or without axillary or internal mammary lymph node involvement
cN3a	Metastasis in ipsilateral infraclavicular lymph node(s)
cN3b	Metastasis in ipsilateral internal mammary lymph node(s) and axillary lymph node(s)
cN3c	Metastasis in ipsilateral supraclavicular lymph node(s)
Note: (sn) and (f) suffixes should be added to the N category to denote confirmation of metastasis by sentinel node biopsy or fine needle aspiration/core needle biopsy, respectively.	
Pathologic (pN)	
pNX	Regional lymph nodes cannot be assessed (for example, previously removed, or not removed for pathologic study)

pN0	No regional lymph node metastasis identified histologically, or isolated tumor cell clusters (ITCs) only. Note: ITCs are defined as small clusters of cells ≤ 0.2 mm, or single tumor cells, or a cluster of < 200 cells in a single histologic cross-section; ITCs may be detected by routine histology or by immunohistochemical (IHC) methods; nodes containing only ITCs are excluded from the total positive node count for purposes of N classification but should be included in the total number of nodes evaluated
pN0(i)	No regional lymph node metastases histologically, negative IHC
pN0(i+)	ITCs only in regional lymph node(s)
pN0(mol-)	No regional lymph node metastases histologically, negative molecular findings (reverse transcriptase polymerase chain reaction [RT-PCR])
pN0(mol+)	Positive molecular findings by RT-PCR; no ITCs detected
pN1	Micrometastases; or metastases in 1-3 axillary lymph nodes and/or in internal mammary nodes; and/or in clinically negative internal mammary nodes with micrometastases or macrometastases by sentinel lymph node biopsy
pN1mi	Micrometastases (200 cells, > 0.2 mm but none > 2.0 mm)
pN1a	Metastases in 1-3 axillary lymph nodes (at least 1 metastasis > 2.0 mm)
pN1b	Metastases in ipsilateral internal mammary lymph nodes, excluding ITCs, detected by sentinel lymph node biopsy
pN1c	Metastases in 1-3 axillary lymph nodes and in internal mammary sentinel nodes (ie, pN1a and pN1b combined)
pN2	Metastases in 4-9 axillary lymph nodes; or positive ipsilateral internal mammary lymph nodes by imaging in the absence of axillary lymph node metastases
pN2a	Metastases in 4-9 axillary lymph nodes (at least 1 tumor deposit > 2.0 mm)

pN2b	Clinically detected* metastases in internal mammary lymph nodes with or without microscopic confirmation; with pathologically negative axillary lymph nodes
pN3	Metastases in ≥ 10 axillary lymph nodes; or in infraclavicular (level III axillary) lymph nodes; or positive ipsilateral internal mammary lymph nodes by imaging in the presence of one or more positive level I, II axillary lymph nodes; or in > 3 axillary lymph nodes and micrometastases or macrometastases by sentinel lymph node biopsy in clinically negative ipsilateral internal mammary lymph nodes; or in ipsilateral supraclavicular lymph nodes
pN3a	Metastases in ≥ 10 axillary lymph nodes (at least 1 tumor deposit > 2.0 mm); or metastases to the infraclavicular (level III axillary lymph) nodes
pN3b	pN1a or pN2a in the presence of cN2b (positive internal mammary nodes by imaging) or pN2a in the presence of pN1b
pN3c	Metastases in ipsilateral supraclavicular lymph nodes
*"Clinically detected" is defined as detected by imaging studies (excluding lymphoscintigraphy) or by clinical examination and having characteristics highly suspicious for malignancy or a presumed pathologic macrometastasis on the basis of FNA biopsy with cytologic examination.	
Distant metastasis (M)	
M0	No clinical or radiographic evidence of distant metastasis
cM0(i+)	No clinical or radiographic evidence of distant metastases in the presence of tumor cells or deposits no larger than 0.2 mm detected microscopically or by molecular techniques in circulating blood, bone marrow, or other nonregional nodal tissue in a patient without symptoms or signs of metastase
cM1	Distant metastases detected by clinical and radiographic means
pM1	Any histologically proven metastases in distant organs; or if in non-regional nodes, metastases > 0.2 mm

Table 2. Histologic Grade (Open Table in a new window)

Histologic grade (G)	
GX	Grade cannot be assessed
G1	Low combined histologic grade (favorable)
G2	Intermediate combined histologic grade (moderately favorable)
G3	High combined histologic grade (unfavorable)

Table 3. Anatomic Stage/Prognostic Groups

(Open Table in a new window)

Stage	T	N	M
0	Tis	N0	M0
IA	T1	N0	M0
IB	T0	N1mi	M0
	T1	N1mi	M0
IIA	T0	N1	M0
	T1	N1	M0
	T2	N0	M0

IIB	T2	N1	M0
	T3	N0	M0
IIIA	T0	N2	M0
	T1	N2	M0
	T2	N2	M0
	T3	N1	M0
	T3	N2	M0
IIIB	T4	N0	M0
	T4	N1	M0
	T4	N2	M0
IIIC	Any T	N3	M0
IV	Any T	Any N	M1

Notes:

- T1 includes T1mi.
- T0 and T1 tumors with nodal micrometastases (N1mi) are staged as Stage IB.
- T2, T3, and T4 tumors with nodal micrometastases (N1mi) are staged using the N1 category.
- M0 includes M0(i+)
- The designation pM0 is not valid; any M0 is clinical.
- If a patient presents with M1 disease prior to neoadjuvant systemic therapy, the stage is considered stage IV and remains stage IV regardless of response to neoadjuvant therapy.
- Stage designation may be changed if postsurgical imaging studies reveal the presence of distant metastases, provided the studies are performed within 4 months of diagnosis in the absence of disease progression, and provided the patient has not received neoadjuvant therapy.

- Staging following neoadjuvant therapy is designated with “yc” or “yp” prefix to the T and N classification. No anatomic stage group is assigned if there is a complete pathologic response (pCR) to neoadjuvant therapy, for example, ypT0ypN0cM0.

Table 4. Prognostic Stage IA (Open Table in a new window)

TNM	Grade	HER2	ER	PR
T1 N0 M0	G1	Positive	Positive	Any
T0 N1 mi, M0	G1	Positive	Negative	Any
T1 N1mi M0	G1 or G2	Negative	Positive	Any
T1 N1mi M0	G1 or G2	Negative	Negative	Positive
T1 N1mi M0	G2 or G3	Positive	Any	Any
T1 N1mi M0	G3	Negative	Positive	Positive

Table 5. Prognostic Stage IB (Open Table in a new window)

TNM	Grade	HER2	ER	PR
T1 N1mi M0	G1 or G2	Negative	Negative	Negative
T0 N1 M0 T1 N1 M0 T2 N0 M0	G1 or G2	Any	Positive	Positive
T0 N1 M0 T1 N1 M0 T2 N0 M0	G3	Positive	Positive	Positive
T2 N1 M0 T3 N0 M0	Any	Positive	Positive	Positive
T1 N1mi M0	G3	Negative	Positive	Negative
T1 N1m, M0	G3	Negative	Negative	Any
T0 N1 M0 T1 N1 M0	G3	Positive	Positive	Positive

T2 N0 M0				

Table 6. Prognostic Stage IIA (Open Table in a new window)

TNM	Grade	HER2	ER	PR
T0 N1 M0 T1 N1 M0 T2 N0 M0	G1 or G2	Positive	Positive	Negative
T0 N1 M0 T1 N1 M0 T2 N0 M0	G1 or G2	Positive	Negative	Any
T0 N1 M0 T1 N1 M0 T2 N0 M0	G1 or G2	Negative	Positive	Negative
T0 N1 M0 T1 N1 M0 T2 N0 M0	G1	Negative	Negative	Any
T0 N1 M0 T1 N1 M0 T2 N0 M0	G2	Negative	Negative	Positive

Table 7. Prognostic Stage IIB (Open Table in a new window)

TNM	Grade	HER2	ER	PR
T0 N1c M0 T1b N1c M0 T2 N0 M0	G2	Negative	Negative	Negative
T0 N1c M0	G3	Negative	Positive	Negative

T1b N1c M0 T2 N0 M0				
T0 N1c M0 T1b N1c M0 T2 N0 M0	G3	Negative	Negative	Any
T2 N1d M0 T3 N0 M0	G1 or G2	Positive	Negative	Negative
T2 N1d M0 T3 N0 M0	G1 or G2	Negative	Positive	Negative
T2 N1d M0 T3 N0 M0	G1 or G2	Negative	Negative	Any
T2 N1d M0 T3 N0 M0	G3	Positive	Positive	Negative
T2 N1d M0 T3 N0 M0	G3	Positive	Negative	Any
T2 N1d M0 T3 N0 M0	G3	Negative	Positive	Positive
T0 N2 M0 T1b N2 M0 T2 N2 M0 T3 N1d M0 T3 N2 M0	G3	Positive	Positive	Positive

Table 8. Prognostic Stage IIIA (Open Table in a new window)

TNM	Grade	HER2	ER	PR
T2 N1d M0 T3 N0 M0	G3	Negative	Positive	Negative
T2 N1d M0 T3 N0 M0	G3	Negative	Negative	Positive
T0 N2 M0 T1b N2 M0 T2 N2 M0 T3 N1d M0 T3 N2 M0	Any	Positive	Positive	Negative
T0 N2 M0 T1b N2 M0 T2 N2 M0 T3 N1d M0 T3, N2, M0	Any	Positive	Negative	Any
T0 N2 M0 T1b N2 M0 T2 N2 M0 T3 N1d M0 T3 N2 M0	G1 or G2	Negative	Positive	Negative
T0 N2 M0 T1b N2 M0 T2 N2 M0 T3 N1d M0 T3 N2 M0	G2	Negative	Negative	Positive
T0 N2 M0	G3	Negative	Positive	Positive

T1b N2 M0 T2 N2 M0 T3 N1d M0 T3 N2 M0				
T4 N0 M0 T4 N1d M0 T4 N2 M0 Any T N3 M0	G1 or G2	Positive	Positive	Positive

Table 9. Prognostic Stage IIIB (Open Table in a new window)

TNM	Grade	HER2	ER	PR
T2 N1d M0 T3 N0 M0	G2 or G3	Negative	Negative	Negative
T0 N2 M0 T1b N2 M0 T2 N2 M0 T3 N1d M0 T3 N2 M0	G1 or G2	Negative	Negative	Negative
T0 N2 M0 T1b N2 M0 T2 N2 M0 T3 N1d M0 T3 N2 M0	G3	Negative	Positive	Negative
T0 N2 M0 T1b N2 M0 T2 N2 M0 T3 N1d M0 T3 N2 M0	G3	Negative	Negative	Positive

T4 N0 M0 T4 N1d M0 T4 N2 M0 Any T N3 M0	G1 or G2	Positive	Positive	Negative
T4 N0 M0 T4 N1d M0 T4 N2 M0 Any T N3 M0	G1 or G2	Positive	Negative	Any
T4 N0 M0 T4 N1d M0 T4 N2 M0 Any T N3 M0	G1 or G2	Negative	Positive	Any
T4 N0 M0 T4 N1d M0 T4 N2 M0 Any T N3 M0	G1 or G2	Negative	Negative	Positive
T4 N0 M0 T4 N1d M0 T4 N2 M0 Any T N3 M0	G3	Positive	Any	Any
T4 N0 M0 T4 N1d M0 T4 N2 M0 Any T N3 M0	G3	Negative	Positive	Positive

Table 10. Prognostic Stage IIIC (Open Table in a new window)

TNM	Grade	HER2	ER	PR

T0 N2 M0 T1b N2 M0 T2 N2 M0 T3 N1d M0 T3 N2 M0	G3	Negative	Negative	Negative
T4 N0 M0 T4 N1d M0 T4 N2 M0 Any T N3 M0	G1 or G2	Negative	Negative	Negative
T4 N0 M0 T4 N1d M0 T4 N2 M0 Any T N3 M0	G3	Negative	Positive	Negative
T4 N0 M0 T4 N1d M0 T4 N2 M0 Any T N3 M0	G3	Negative	Negative	Any

Table 11. Prognostic Stage IV (Open Table in a new window)

TNM	Grade	HER2	ER	PR
Any T, Any N, M1	Any	Any	Any	Any

Questions & Answers

Overview

How is breast cancer staged?

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